

Manish Nalumachu

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SKILLS

Programming: Python, C++, SQL, Bash, Linux
Tools & DevOps: Git, GitHub, VS Code, CMake, Docker, Apache, Postman, Splunk, Grafana
Cloud & Infrastructure: AWS, GCP, Amazon EC2, Firebase
Data & Visualization: NumPy, Pandas, Matplotlib, yfinance, Google Colab
Backend & APIs: FastAPI, GraphQL, Kafka, Redis, MongoDB, Postman
Databases: Firebase, MongoDB, Redis, Neo4j, Supabase, FAISS, ChromaDB, Vector Databases
AI/ML Frameworks: LangChain, LangGraph, MCP Server, PEFT (LoRA/QLoRA), RAG (Retrieval-Augmented Generation)
LLMs & AI Tools: OpenAI API, Claude, Hugging Face, Cline
Soft Skills: Problem Solving, Self-Learning, Adaptability, Leadership, Verbal Communication, Vendor Management

EDUCATION

Degree	Institution	CGPA/%	Year
B.Tech.	JNTUH College of Engineering	7.5	2022–2026
HSC (XII)	Alphores Junior College	93%	2020–2022
SSC (X)	Paramita High School	100%	2020

ACHIEVEMENTS & CERTIFICATIONS

- **IMC Prosperity 4 Global Trading Competition (Apr 2026)** — Ranked **#4479** / **~9,000+** teams globally; built Python algorithmic trading bots exploiting a deterministic +1,000 tick/day price trend; implemented fixed fair-value market making with nonlinear inventory skew across two instruments.
- **LeetCode** — Solved **1,100+** DSA problems; **Top 0.3%** globally; maintained a **180-day coding streak** demonstrating consistent discipline and algorithm mastery. [\[Profile\]](#)
- **15+ personal projects** spanning AI agents, LLM research, quant systems, full-stack apps, and systems programming.
- **Certificates** — [Google Drive](#)

CORE COURSEWORK

Data Structures & Algorithms · System Design · OOP · DBMS · Operating Systems · Microservices · Computer Networks · LLD

EXPERIENCE

Full-Stack Developer Intern May – Aug 2024
SWIZOSOFT (OPC) PRIVATE LIMITED Remote

- Built **TaskFlow**, a full-stack task management app using **Python Flask, SQLite, SQLAlchemy** with JWT auth, full CRUD, priority levels, due-date tracking, and an analytics dashboard; handled ~500 concurrent task records.
- Designed 8 RESTful API endpoints; reduced query response time by ~40% via indexed DB schema and connection pooling.

PERSONAL PROJECTS (15+ total — selected highlights)

Quant Backtesting Engine [GitHub](#)

Event-driven system for testing trading strategies on historical market data with realistic profit and risk evaluation.

- Built an **event-driven engine** that processes price data, generates buy/sell signals, executes trades, and computes overall **profit/loss** for a strategy.
- Supports **momentum and mean-reversion strategies** with a modular design for adding custom strategies.
- Computes key metrics including **CAGR, Sharpe Ratio (~1.0–1.5)**, max drawdown (~10–20%), and win-rate (~55–60%) to evaluate performance and risk.
- Incorporated realistic trading factors such as **slippage (0.1%)** and commissions for near real-world simulation.
- **Results:** Handles **100K+ data points**, runs a full 1-year backtest in **<2 seconds**, and produces stable, realistic profit.
- **Stack:** Python, Pandas, NumPy, Matplotlib, yfinance

Mini-LLM [GitHub](#)

GPT-style transformer built to understand LLM internals — from tokenisation to autoregressive text generation.

- Built a GPT-style Transformer that reads text, learns patterns using **multi-head self-attention** ($h=8$, $d_{\text{model}}=512$), and generates new text by predicting the next token (**~6M parameters**).
- Trained on a character-level dataset; achieves **validation loss of ~1.4** after **10K training steps** and generates coherent sequences up to **200 tokens**.
- Applied **LoRA fine-tuning** (rank=8, $\alpha=16$) to reduce trainable parameters by ~94
- **Stack:** Python, PyTorch, NumPy, Hugging Face Tokenizers

Fitness Tracker — NutraX [GitHub](#)

AI-powered fitness & nutrition platform with real-time biometrics, smart notifications, and a production-grade backend.

- Built **FastAPI backend with Redis caching** (TTL=300s) for session management and high-frequency biometric writes; reduced average DB read latency from **120ms → 18ms** (85% improvement).
- Integrated **Gemini API chatbot** with conversation context window (last 10 messages) for personalised fitness advice; **ML Kit pose detection** runs at **30 FPS** for real-time workout form analysis and per-set calorie estimation.
- Developed **Firebase Auth + Firestore** with real-time sync across devices, push notifications, in-app gallery, and notes; supports **~1,000 concurrent users** on free tier via efficient listener management.
- **Stack:** Flutter, FastAPI, Redis, Firebase, Gemini API, ML Kit

EXTRACURRICULAR ACTIVITIES

- **Competitive:** [LeetCode](#) · [Codeforces](#) · [Chess.com](#) · [Tensor Tonic](#)
- **Languages:** English, Hindi, Telugu
- **Hobbies:** Gym, Guitar
- **Blogs:** I write blogs
- **Comic Writer:** Created and wrote original comic stories and scripts

Medium
Tapas